

KOTZEBUE, AK, USA

WMO: 701330

Lat: 66.886N

Lon: 162.613W

Elev: 9

StdP: 101.22

Time Zone: -9.00 (NAK)

Period: 94-19

WBAN: 26616

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-36.1	-33.7	-36.8	0.1	-33.1	-34.9	0.1	-31.3	17.7	-10.2	16.1	-10.9	2.2	60	0.931

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	4.8	20.5	15.6	18.5	14.4	17.1	13.7	16.2	19.6	15.1	18.1	14.1	16.6	4.3	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.7	10.4	18.0	13.5	9.7	16.8	12.5	9.0	15.8	45.3	19.5	42.3	18.0	39.5	16.6	21.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
15.3	13.1	11.7	DB	-37.7	23.7	3.5	2.4	-40.2	25.4	-42.3	26.8	-44.3	28.1	-46.8	29.9	(4)
			WB	-35.4	17.5	2.5	1.7	-37.2	18.7	-38.6	19.7	-40.0	20.7	-41.8	22.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-4.2	-18.6	-16.4	-16.6	-8.6	0.7	8.6	13.0	11.3	6.5	-2.8	-11.6	-16.3
	DBStd	13.09	9.62	8.83	7.46	6.61	4.98	4.57	3.35	3.15	3.61	5.30	6.79	7.58
	HDD10.0	5383	887	738	824	557	289	79	11	21	114	397	649	816
	HDD18.3	8216	1146	972	1083	807	546	292	169	217	356	655	899	1074
	CDD10.0	213	0	0	0	0	2	38	103	63	8	0	0	0
	CDD18.3	4	0	0	0	0	0	1	3	1	0	0	0	0
	CDH23.3	7	0	0	0	0	0	3	4	0	0	0	0	0
	CDH26.7	1	0	0	0	0	0	1	0	0	0	0	0	0
Wind	WSAvg	5.1	5.2	5.6	4.6	4.7	4.3	4.7	4.9	5.2	5.2	5.6	5.7	5.3
Precipitation	PrecAvg	248	14	12	10	11	10	14	39	51	39	22	16	16
	PrecMax	442	45	31	56	47	35	45	89	134	110	61	57	48
	PrecMin	132	0	0	0	1	0	0	0	2	1	0	0	3
	PrecStd	66	11	8	10	10	8	10	21	32	23	14	13	10
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.8	1.4	1.8	4.8	15.6	22.6	23.8	21.7	16.0	8.3	2.3	1.1
		MCWB	0.7	0.4	0.7	2.4	9.5	16.1	17.0	16.2	11.2	4.6	1.1	0.3
	2%	DB	-0.3	0.0	-0.8	3.0	11.5	19.3	21.2	18.8	13.9	6.2	1.0	-0.7
		MCWB	-1.2	-0.8	-1.6	1.0	7.4	14.3	15.9	14.8	10.6	4.3	0.3	-1.6
	5%	DB	-2.7	-1.8	-3.2	1.8	8.8	17.3	19.1	17.2	12.4	4.9	-0.2	-3.0
		MCWB	-3.4	-2.5	-4.0	0.5	5.8	13.3	15.1	14.1	9.9	3.6	-0.9	-3.8
	10%	DB	-4.8	-4.3	-5.5	0.9	6.6	15.2	17.6	15.9	11.2	3.6	-2.0	-5.6
		MCWB	-5.5	-5.0	-6.1	0.0	4.4	11.7	14.3	13.2	9.2	2.5	-2.8	-6.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.8	0.7	0.8	2.6	9.9	17.1	18.0	17.3	12.2	6.0	1.3	0.5
		MCDB	1.7	1.2	1.7	4.6	14.6	21.2	22.1	20.1	14.6	6.9	1.9	1.0
	2%	WB	-1.0	-0.7	-1.5	1.3	7.9	14.9	16.6	15.6	11.2	4.7	0.4	-1.5
		MCDB	-0.2	-0.2	-0.8	2.5	11.3	18.5	20.1	18.2	13.1	5.7	1.0	-0.8
	5%	WB	-3.2	-2.4	-3.9	0.7	6.0	13.4	15.5	14.5	10.3	3.7	-0.8	-3.7
		MCDB	-2.5	-1.7	-3.2	1.7	8.5	16.8	18.7	16.8	12.0	4.8	-0.2	-2.9
	10%	WB	-5.2	-4.8	-6.2	-0.1	4.5	12.0	14.6	13.5	9.6	2.4	-2.7	-6.1
		MCDB	-4.5	-4.2	-5.5	1.0	6.6	15.1	17.3	15.6	11.1	3.6	-1.9	-5.6
Mean Daily Temperature Range	5% DB	MDBR	6.4	7.0	7.4	7.5	5.8	5.6	4.8	4.4	4.4	3.9	4.9	6.1
		MCDBR	6.8	7.6	7.2	6.3	8.8	7.9	7.0	6.3	5.6	3.8	4.2	5.7
		MCWBR	6.5	7.4	6.8	5.3	5.6	5.0	4.2	3.8	3.5	3.0	4.1	5.5
	5% WB	MCDBR	7.0	7.4	7.3	6.0	8.3	7.0	6.2	5.5	4.7	3.5	4.1	6.1
		MCWBR	6.7	7.3	7.0	5.2	5.5	4.8	4.2	3.6	3.1	3.0	4.0	5.9
Clear-Sky Solar Irradiance	taub	0.172	0.238	0.257	0.319	0.357	0.338	0.350	0.360	0.304	0.254	N/A	N/A	
	taud	1.899	2.111	2.224	2.083	2.106	2.399	2.465	2.459	2.547	2.377	N/A	N/A	
	Ebn at Noon	211	608	810	825	829	868	845	793	770	613	N/A	0	
	Edh at Noon	14	50	77	118	130	100	91	83	60	39	N/A	0	
All-Sky Solar Radiation	RadAvg	0.07	0.60	2.36	4.44	5.75	6.22	4.82	3.51	2.17	0.78	0.14	0.00	
	RadStd	0.01	0.08	0.20	0.30	0.54	0.45	0.44	0.44	0.23	0.09	0.02	0.00	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		+1.43	N/A	N/A	+1.00	+0.82	N/A	-475	-510	+65	+1
Regional (8 neighbors)		+1.43	N/A	N/A	+1.00	+0.82	N/A	-475	-510	+65	+1

Nomenclature: See separate page